

Dashboards to Decisions

A Self-Correcting ML Loop with Elastic Observability & Analytics

Watch. **Retrain.** Ship. A custom ML evidence



Need thy help!

1

Do you have a model in production right now?

2

Would you know within a while that it stopped being good?

What usually happens

QUESTION 1 : A MODEL IN PRODUCTION



QUESTION 2 : WOULD YOU KNOW?



Going forward,

79,491

credentials having names and descriptions,
free text

0

ground-truth labels to grade the grouping
against

0

alerts when it drifts, because there is nothing to
alert on

What is actually in this model?

UNIVERSITY

Bachelor of Science in Computer Science

BOOTCAMP

Full-Stack Web Development Certificate

VENDOR

Cloud Practitioner Badge

What we have

About ~79K credentials!

University degrees, community college certificates, apprenticeships, two-hour online badges.

What the model does

It reads the text of each credential, turns the words into numbers, and puts similar ones in the same pile. Nobody tells it what the piles should be, or how many. Five strategies compete and the best metric scoring is chosen.

Why it is hard

You cannot score it by checking answers. The inter and intra distance between them vary with every incoming batch of data.

The five methods

M1

Weighted Cosine

merge the closest pairs

M2

Binary Jaccard

shared words, not
weights

M3

**Spherical
KMeans**

k piles, elbow-picked

M4

HDBSCAN

finds crowds, may
refuse

M5

Two-Stage

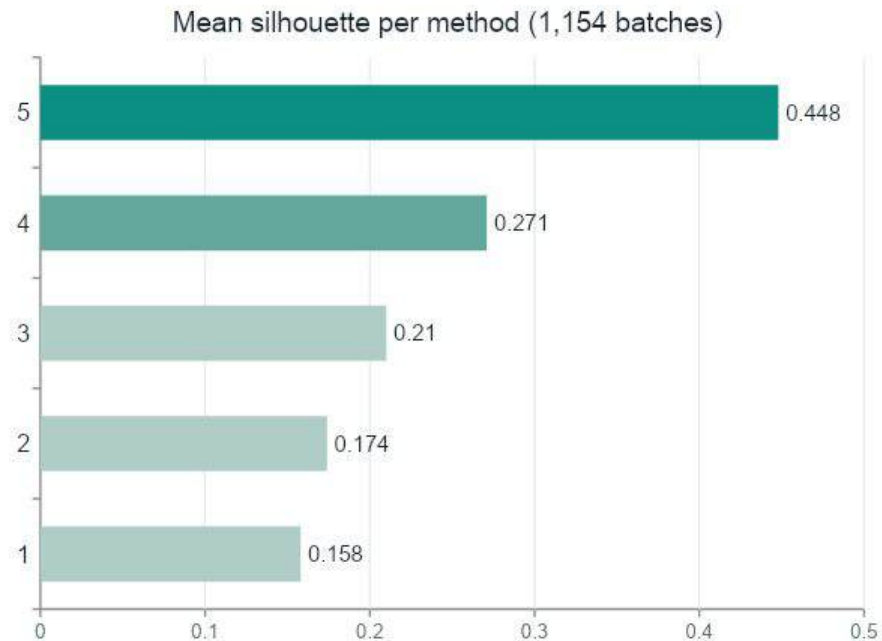
rough sort, then merge

Factors derived from Batches and Methods

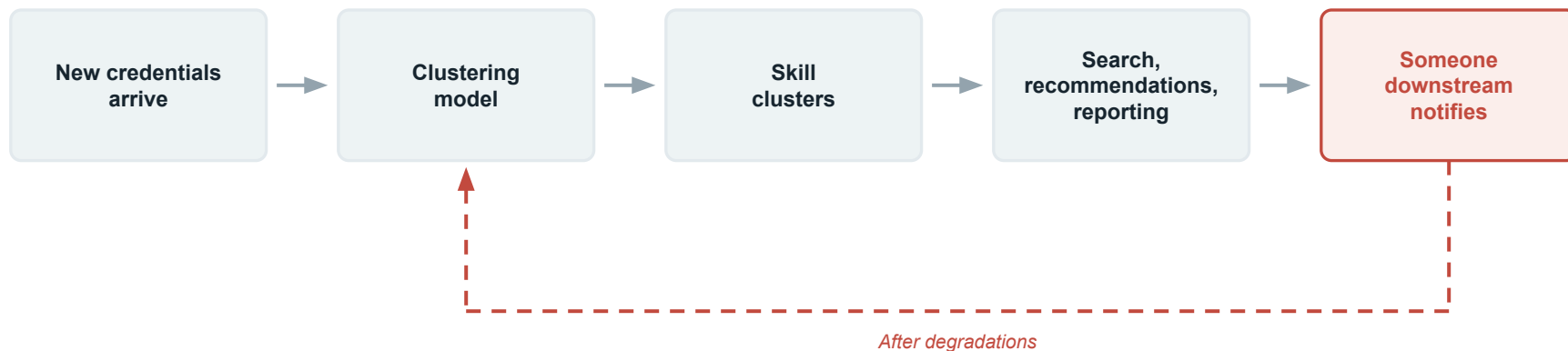
Mean silhouette is the vital sign

Also, some other metrics are explored as deciding factors:

1. Population Stability Index (Has the vocabulary shifted? Ex: From Academic to Vendor based)
2. Type Mix shift (Type of Drift)



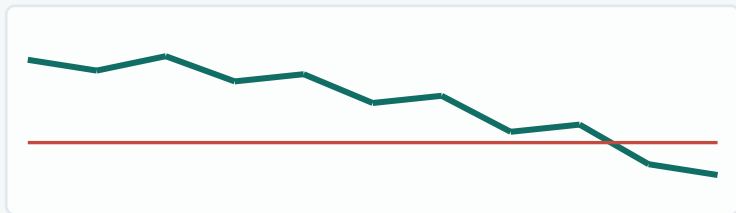
The failure is silent. The feedback loop is a human.



Differences between monitoring mechanisms

Supervised

accuracy

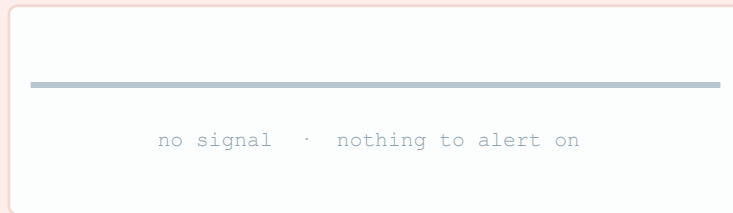


ALERT · accuracy below threshold

- There is a right answer, so the model can be graded.
- The number moves, the alert fires, someone acts.

Unsupervised (this pipeline)

accuracy



SILENT · and the clusters are drifting anyway

- No labels, so nothing to grade and no threshold to breach.

Monitoring the vital signs is the key.

Why Elastic Observability/Analytics?



- A patient who cannot tell you what hurts.
- So you watch vital signs, continuously.
- You never act on one bad reading and analyse the trend.
- That trend rule is the retrain gate.

Why Elastic?

Multiple Information

Heterogenous document storage (ML metrics/Batch Events/ Records showing failures)

Decision Making

Before any promotion of version, Python Loop reads recent batch health from ml-events

ES|QL supports live aggregation

No need to maintain separate summary tables

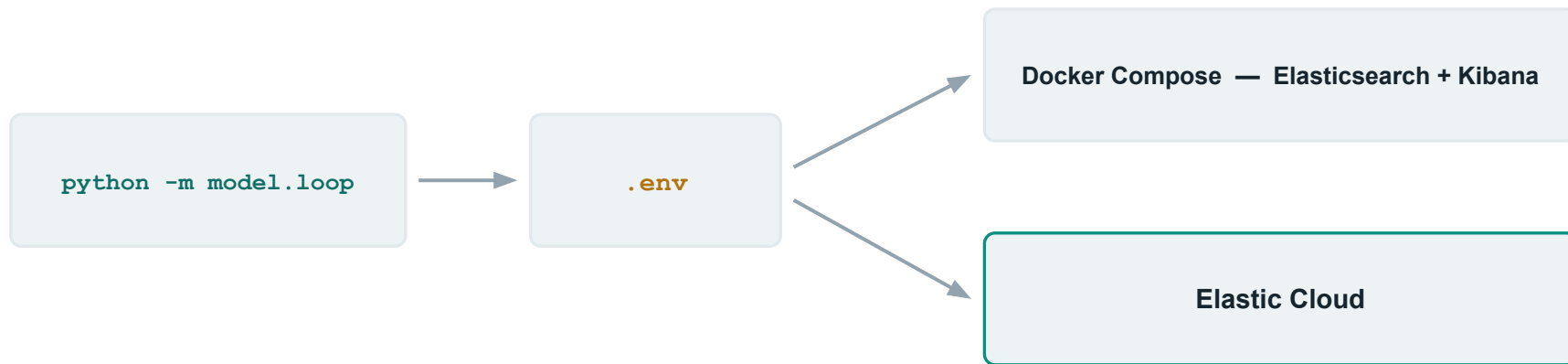
Supports structured data and free text

It has dynamic acceptance/search of types of data

One client

the same code points at local Docker or Elastic Cloud

Local and Cloud: the same model, the same loop



Demo

```
$ python -m model.loop #to run the new records
```

01

The per-method silhouette panel

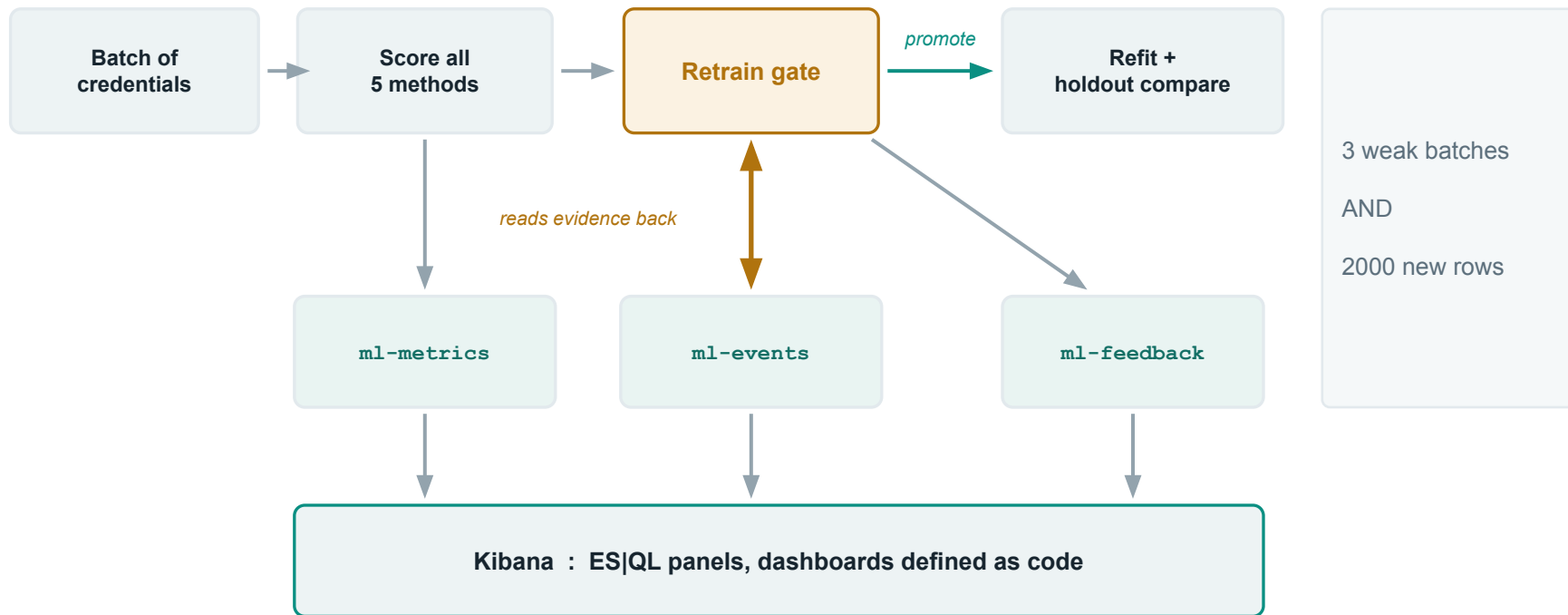
02

The gate event

03

The version record

Where each Elastic piece actually sits



What makes it self-correcting: the two-key rule

```
model/loop.py
```

```
can_retrain = weak_count >= MIN_CONSECUTIVE and new_since_fit >= MIN_NEW_SAMPLES    # key 1: local  
  
window      = sink.recent_batch_health(serving_name, limit=WEAK_LOOKBACK)           # ask Elastic  
  
can_retrain = len(evidence_batches) >= MIN_CONSECUTIVE                             # key 2: Elastic can veto
```

1 loop.py proposes

Three weak readings in the last eight,
and enough new rows since the last fit.



BOTH KEYS TURN → v4 SHIPS

```
evidence_source: elasticsearch  
evidence_batches: [01035, 01036, 01037]  
previous: v3  ->  new_version: v4
```



2 Elastic confirms

The stored batch health has to show
the same thing or the promotion is
held.

Turn one key only and nothing ships! - Can be ALTERED!

Would you retrain this model?

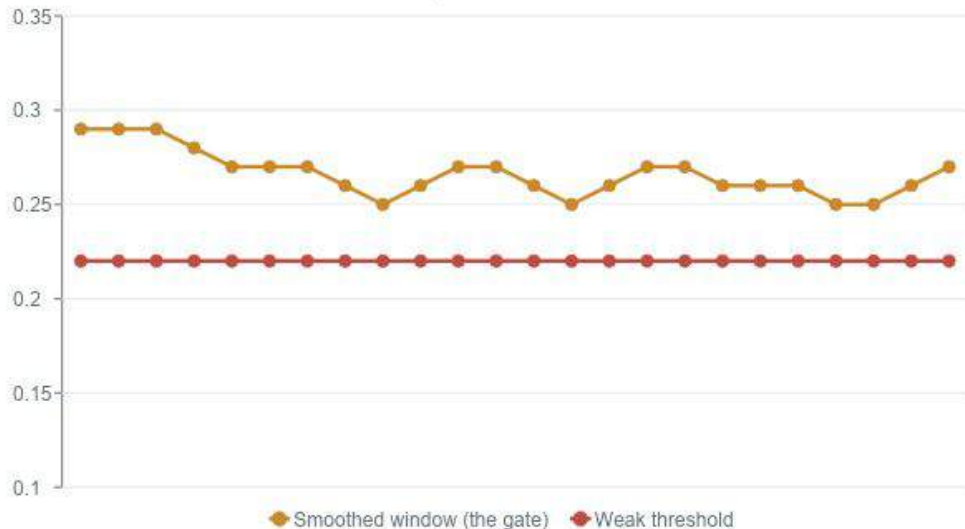
THE RULE

Retrain after three consecutive weak readings, measured on a 400-row smoothed window.

A Retrain it

B Looks good

Mean silhouette, 24 consecutive batches



What the gate could not see

Smoothing averages several batches together, so the line looks healthier than any single batch really was.

Retraining should react to the current state!

Fix: Remember the 2 gates: Quality and Quantity



Retraining after this Integration

01

How Retraining happens?

All five methods are retrained on the original data plus recent rows, then scored on the same last batch credentials and the current production model is scored there too, so you get a real before-and-after.

If the best new model is more than 0.01 worse, the retrain is rejected and production stays

If another method wins by more than 0.02, serving switches, otherwise the same method ships with fresh parameters.

02

Prevent Loops

As the newer versions are produced, the data and the weak batch window is cleared.

On rejection the weak window is cleared too. Incase of new entries in the weak window the counter is set to begin again.

```
v4 → batch-01035 ... batch-01037, 2,000 rows
```



Three words to take away

Watch. Retrain. Ship.

If the model can't tell you it's wrong, watch its vitals, and don't ship anything the data won't back up.

Watch

Every batch, every method

Retrain

The gate asks Elasticsearch before it promotes or an alert fires it.

Ship

The version records the exact batches that justified it.

<https://github.com/kritikaberry/Self-Correcting-ML-Loop-with-Elastic-Observability?tab=readme-ov-file>

Let's connect!



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Let's connect!

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